

MOUNAH HONORÉ CÉSAIRE

System & Performance Engineer

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SUMMARY

System & performance engineer and finishing Ph.D. student (expected in Jan 2026) in Operating Systems with hands-on experience in **Linux kernel development**, including **real contributions to VPN performance and CPU energy efficiency**. Skilled in kernel networking, cloud-native deployments, and performance optimization. A problem-solver with strong teamwork abilities, dedicated to tackling complex, impactful challenges through rigorous engineering and innovation.

WORK EXPERIENCE

Ph.D: Understanding Linux Scheduling Bottlenecks

Inria Rennes

📅 December 2022 – Ongoing 📍 Rennes, France

Improving the Performance of a Kernel VPN: The Case of WireGuard:

- Investigated performance bottlenecks in the WireGuard VPN module, identifying Execution Order Inversion (Eol) as the root cause of a 5× throughput drop
- Designed and implemented asynchronous kernel APIs (kernel threads,workqueues) to resolve the bottleneck and bring a 5× improvement in terms of throughput and 2× improvement in terms of latency.

Assessing the Energy Efficiency of Intel Heterogeneous CPUs:

- Built an automated benchmarking framework to evaluate Intel heterogeneous CPUs (P-cores/E-cores) across DVFS governors and workload types.
- Delivered actionable insights for scheduler design and CPU tuning, balancing performance and energy efficiency

Research Internship

IRISA

📅 June 2022 – September 2022 📍 Rennes, France

Cloud-Native Malware Detection Platform on Serverless:

- Designed and deployed a serverless detection pipeline using OpenFaaS and OpenWhisk, enabling on-demand analysis of Android network traffic.
- Benchmarked and compared FaaS-based detection performance with traditional anti-malware solutions.

PUBLICATIONS

👥 Conference Proceedings

- H. C. Mounah, D. Mvondo, J. Lawall, and Y.-D. Bromberg, “The impact of kernel asynchronous apis on the performance of a kernel vpn,” in *Proceedings of the 18th ACM International Systems and Storage Conference*, ser. SYSTOR '25, Virtual, Israel: Association for Computing Machinery, 2025, pp. 167–173, ISBN: 9798400721199. DOI: 10.1145/3757347.3759133.

EDUCATION

Ph.D in Computer Science

University of Rennes

📅 Dec 2022 – Ongoing

Engineering Degree in Computer Science

National Advanced School of Engineering of Yaounde, Cameroon

📅 Sept 2017 – Oct 2022

TECHNICAL SKILLS

Low Level Languages: C, Go

Scripting: Python, Bash JavaScript

Performance and Optimization: Perf, eBPF

Dev Tools: Git, Gitlab CI, Github Actions

DevOps: Ansible, Docker, Kubernetes

SOFT SKILLS

- Teamwork and Collaboration
- Problem Solving
- Communication
- Critical Thinking

LANGUAGES

- French: Native
- English: B2 (Fluent in reading and writing)

INTERESTS

- Music
- Reading